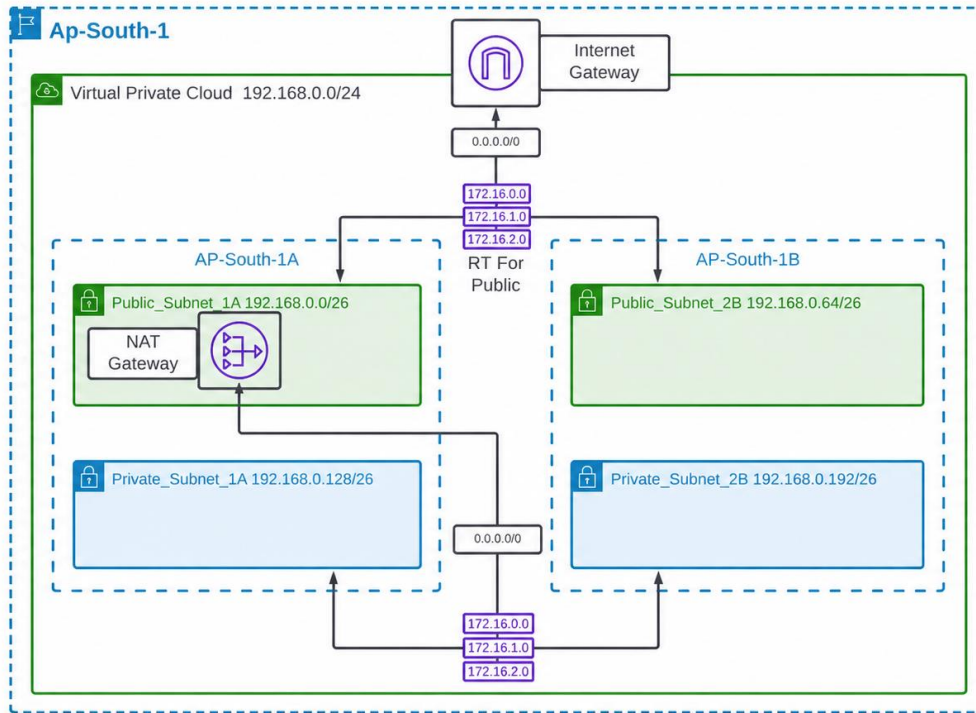


Network Load Balancer

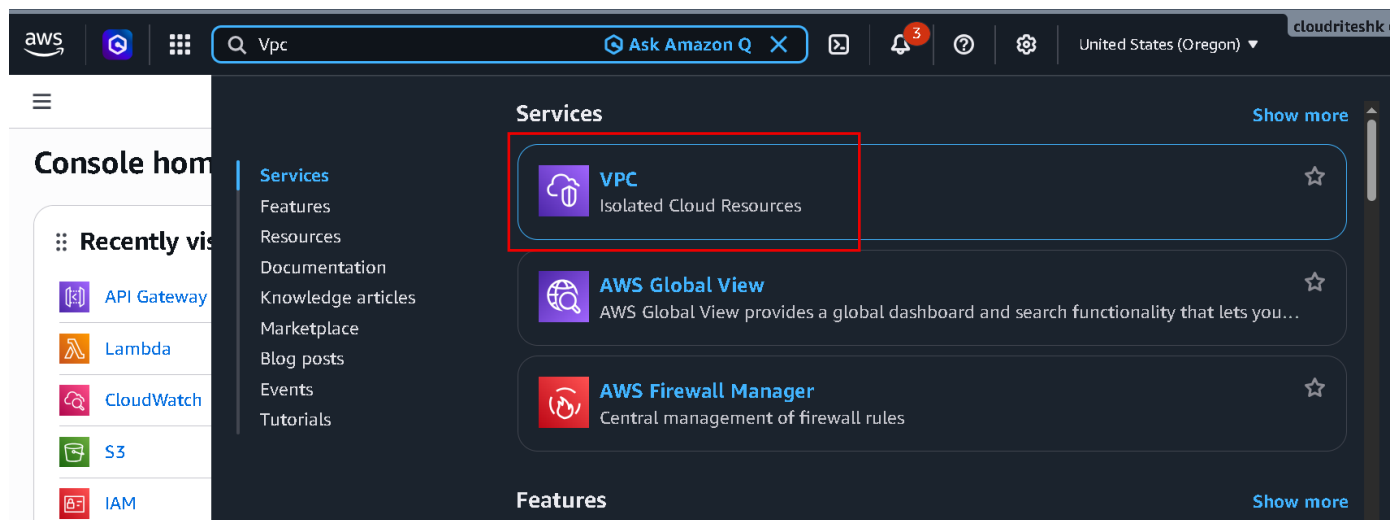
Summary: -

I worked on a Network Load Balancer where incoming traffic was distributed between multiple servers to improve speed, handle high traffic, and keep the application running smoothly. I also configured networking, security, and server connectivity for better performance and availability.

Implementation Load Balancer VPC Design: -



- Firstly, we create our VPC.
- Go to your AWS Console.



- Now Search VPC.
- And open it VPC.

VPC dashboard

AWS Global View

Filter by VPC:

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways

Create VPC **Launch EC2 Instances**

Note: Your Instances will launch in the United States region.

Resources by Region Refresh Resources

You are using the following Amazon VPC resources

Resource	Count
VPCs	Oregon 1
Subnets	Oregon 2
Endpoint Services	Oregon 0
NAT Gateways	Oregon 0

- Now simply click Create VPC.

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

MyVPC

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

192.168.0.0/24

CIDR block size must be between /16 and /28.

- Select VPC only.
- Then Enter your VPC name.
- And then Enter IPv4 CIDR as 192.168.0.0/24

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q MyVPC X Remove tag

Add tag

You can add 49 more tags

Cancel Preview code **Create VPC**

- Now simply scroll down and click Create VPC.
- Now we need to create four Subnet.
- We are creating two public subnets (public subnets having inbound internet access allowed).
- And we create two private subnets (which have no inbound internet access allowed).

☰ VPC > Your VPCs > vpc-03279cdb550dee663 ⓘ 🖨

VPC dashboard <

AWS Global View ↗

Filter by VPC: ▾

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

DHCP option sets

vpc-03279cdb550dee663 / MyVPC Actions ▾

Details Info

VPC ID
vpc-03279cdb550dee663

DNS resolution
Enabled

Main network ACL
acl-0f2c2865119107247

IPv6 CIDR (Network border group)
-

State
Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Block Public Access
Off

DHCP option set
dopt-04c9fec64e7acfa5a

IPv4 CIDR
192.168.0.0/24

Route 53 Resolver DNS Firewall rule groups
-

DNS hostnames
Disabled

Main route table
rtb-0457fe28e5ad5cea7

IPv6 pool
-

Owner ID
463646775279

- Now click on subnets.

Subnets (4) Info

Last updated 1 minute ago ⌂ Actions ▾ Create subnet

🔍 Find subnets by attribute or tag

☐	Name ▾	Subnet ID ▾	State ▾	VPC ▾
☐	Subnet1(Public)	subnet-052ee09f56668d6c1	Available	vpc-0862eb08d50c280e6
☐	-	subnet-04ffa8e5e9c79ad3d	Available	vpc-072c4acd761c3b942
☐	Subnet1(Private)	subnet-0508623c58480124a	Available	vpc-0862eb08d50c280e6
☐	-	subnet-0bc545e3030c41ef4	Available	vpc-072c4acd761c3b942

Select a subnet

- Now simply click on Create subnet.
- Firstly, we create two public subnets.

Create subnet Info

VPC

VPC ID
Create subnets in this VPC.

vpc-03279cdb550dee663 (MyVPC) ▾

Associated VPC CIDRs

IPv4 CIDRs
192.168.0.0/24

- Select your VPC that you created now.

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public_Subnet_1A

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-s1-az1 (ap-south-1a)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.0/26

64 IPs

- Enter Subnet name as Public_Subnet_1A
- The select Availability Zone (ap-south-1a).
- Then IPv4 subnet CIDR block as 192.168.0.0/26
- Now simply scroll down.

▼ Tags - optional

Key

Q Name



Value - optional

Q Public_Subnet_1A



Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

- Now we are going to create another click Add new Subnet.

Subnet 2 of 2

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public_Subnet_2B

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-s1-az3 (ap-south-1b)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.64/26

64 IPs

- Enter subnet 2 name as Public_Subnet_2B
- Select Availability Zone (ap-south-1b)
- Then Enter IPv4 subnet CIDR block as 192.168.0.64/26
- Now simply scroll down.

▼ Tags - optional

Key

You can add 49 more tags.

Value - optional

- Now click Add new subnet.
- Now we will add Private Subnets.

Subnet 3 of 3

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block

64 IPs

- Enter Subnet name as Private_Subnet_1A
- The select Availability Zone (ap-south-1a).
- Then IPv4 subnet CIDR block as 192.168.0.128/26
- Now simply scroll down.

▼ Tags - optional

Key

You can add 49 more tags.

Value - optional

- Now we are going to create another click Add new Subnet.

Subnet 4 of 4

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private_Subnet_2B

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.192/26

64 IPs

- Enter Subnet name as Private_Subnet_2B
- The select Availability Zone (ap-south-1b).
- Then IPv4 subnet CIDR block as 192.168.0.192/26
- Now simply scroll down.

▼ Tags - optional

Key

Q Name

Value - optional

Q Private_Subnet_2B

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

- Now click Create subnet.

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Subnets (4) [Info](#)

Last updated
5 minutes ago

Actions

Create subnet

Find subnets by attribute or tag

Subnet ID : subnet-0e7823bb0276394a2

Subnet ID : subnet-0eeb1e9dd70507728

Subnet ID : subnet-0f82985187f3f7b70

(+1)

Clear filters

☐	Name	Subnet ID	State	VPC
☐	Public_Subnet_1A	subnet-0e7823bb0276394a2	Available	vpc-03279cdb550dee663 M
☐	Private_Subnet_2B	subnet-0c053b6393c3e8c5f	Available	vpc-03279cdb550dee663 M
☐	Private_Subnet_1A	subnet-0f82985187f3f7b70	Available	vpc-03279cdb550dee663 M
☐	Public_Subnet_2B	subnet-0eeb1e9dd70507728	Available	vpc-03279cdb550dee663 M

- Now our Four Subnet is created.
- Now to provide Internet connectivity, we have to add Internet gateway and we have to attach internet gateway with our VPC.
- Now click Internet gateways.

Internet gateways (2) [Info](#) Actions Create internet gateway

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID
------	---------------------	-------	--------

- Now click Create internet gateway.

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

IG

- Now Enter Name (IG).

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Name

Value - optional

IG

Remove

Add new tag

You can add 49 more tags.

Cancel

Create internet gateway

- Now simply click Create internet gateway.

igw-050a2613d0c562441 / IG

Details [Info](#)

Internet gateway ID

igw-050a2613d0c562441

State

Detached

VPC ID

-

Owner

4636

Actions

Attach to VPC

Detach from VPC

Manage tags

Delete

Tags (1)

Search tags

Manage tags

Key	Value
Name	IG

- Now our internet gateway is created.
- But you see it's not Attach with our VPC.
- Now click Action and then select Attach to VPC.

Attach to VPC (igw-050a2613d0c562441) [Info](#)

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

vpc-03279cdb550dee663

AWS Command Line Interface command

Cancel

Attach internet gateway

- Now Select Create VPC and then Click Attach internet gateway.

VPC > Internet gateways > igw-050a2613d0c562441

igw-050a2613d0c562441 / IG

Details [Info](#)

Internet gateway ID: [igw-050a2613d0c562441](#)

State: **Attached**

VPC ID: [vpc-03279cdb550dee663](#) [MyVPC](#)

Owner: [463646775279](#)

Tags (1) [Manage tags](#)

Search tags

Key	Value
Name	IG

- it's Attach to VPC.
- Now we have to set up route table.
- Now we have a main route table which is default and we want to provide inbound connectivity to this public subnet.
- So, for this I am going to create Separate Route table for public subnet
- Now click on Route tables in nav bar.

Route tables (1/5) [Info](#) Last updated 1 minute ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations
☒	Edit Name Main	0457fe28e5ad5cea7	–	–
☐	–	017299725a38d5666	2 subnets	–
☐	–	0aee4d9128c0e3ce0	–	–
☐	vpc1-	0e19b34caf5643c37	subnet-0508623c584801...	–
☐	vpc1-	res-0bad251f343575c0f	subnet-052ee09f56668d...	–

- Now 1st one is our Default Route table we name it as Main and save it.
- we know that this is our VPC default Route table by clicking Route table ID.
- Now click on Create route table.

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

RT For Public

VPC
The VPC to use for this route table.

vpc-03279cdb550dee663 (MyVPC)

- Enter name of Route table (RT For Public)
- Now select VPC that you created now.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q RT For Public X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

- Now simply scroll down and click Create route table.

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

rtb-0b72240aa9b00dfea / RT For Public

Actions

Details Info

Route table ID

rtb-0b72240aa9b00dfea

Main

No

Explicit subnet associations

–

Edge associations

–

VPC

vpc-03279cdb550dee663 | MyVPC

Owner ID

463646775279

Routes

Subnet associations

Edge associations

Route propagation

Tags

- Now our route table is created.
- But right now, your route table is not associated with public subnets.
- Now click on Route table.

Route tables (1/6) Info

Last updated 13 minutes ago

Actions

Create route table

Find route tables by attribute or tag

< 1 > ⚙

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	Main	rtb-0457fe28e5ad5cea7	–	–	Yes	vpc-03279cdb550
☒	RT For Public	rtb-0b72240aa9b00dfea	–	–	No	vpc-03279cdb550
☐	–	rtb-017299725a38d5666	2 subnets	–	Yes	vpc-072c4acd761
☐	–	rtb-0aee4d9128c0e3ce0	–	–	Yes	vpc-0862eb08d5C

rtb-0b72240aa9b00dfea / RT For Public

⚙

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (0)

Edit subnet associations

- Now select RT For Public route table.
- And click Subnet associations.
- And then click Edit subnet association.

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public_Subnet_1A	subnet-0e7823bb0276394...	192.168.0.0/26	-	Main (rtb-0457fe28e5ad5cea7 /
☐	Private_Subnet_2B	subnet-0c053b6393c3e8c5f	192.168.0.192/26	-	Main (rtb-0457fe28e5ad5cea7 /
☐	Private_Subnet_1A	subnet-0f82985187f3f7b70	192.168.0.128/26	-	Main (rtb-0457fe28e5ad5cea7 /
☒	Public_Subnet_2B	subnet-0eeb1e9dd705077...	192.168.0.64/26	-	Main (rtb-0457fe28e5ad5cea7 /

Selected subnets

subnet-0eeb1e9dd70507728 / Public_Subnet_2B X subnet-0e7823bb0276394a2 / Public_Subnet_1A X

Cancel Save associations

- Now select both public Subnets.
- And click Save association.
- Now we have to add this particular route towards the internet gateway into the route public.

You have successfully updated subnet associations for rtb-0b72240aa9b00dfea / RT For Public.

Route tables (1/6) Info Last updated 3 minutes ago Actions Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	Main	rtb-0457fe28e5ad5cea7	-	-	Yes	vpc-03279cdb550
☒	RT For Public	rtb-0b72240aa9b00dfea	2 subnets	-	No	vpc-03279cdb550
☐	-	rtb-017299725a38d5666	2 subnets	-	Yes	vpc-072c4acd761

rtb-0b72240aa9b00dfea / RT For Public

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (1) Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin

- Now click Routes and then click Edit routes.

Route 2

Destination 0.0.0.0/0 X

Propagated No

Target Internet Gateway X

igw-050a2613d0c562441 X

Route Origin CreateRoute

Remove

Add route

Cancel Preview Save changes

- Now Firstly, click Add route and then Enter 2nd Route details.
- Select Destination 0.0.0.0/0
- Select Target Internet Gateway
- Then select Created Internet Gateway
- And simply click Save Changes.
- Now this is done and we need to Setup NAT Gateway.

- Because you will place your Ec2 instance into the private subnet.
- If we are not going to set up NAT Gateway then you cannot provide outbound internet access to your EC2 instance.
- It means you cannot update your Ec2 instance.
- So, we have to create this NAT Gateway inside the public subnet.

The screenshot shows the AWS VPC console with the 'Route tables' section selected. The specific route table 'rtb-0b72240aa9b00dfea / RT For Public' is displayed. The 'Details' tab is active, showing the Route table ID, VPC (vpc-03279cdb550dee663), Main status (No), Owner ID (463646775279), Explicit subnet associations (2 subnets), and Edge associations (none). The 'Routes' tab is also visible at the bottom.

- Now click NAT Gateway in nav bar.

The screenshot shows the AWS NAT Gateways console. The 'NAT gateways (1)' section is active. A search bar is present. The 'Create NAT gateway' button is highlighted in orange. Below the search bar, a table lists the NAT gateways. One gateway is shown: 'vpc1-nat' with ID 'nat-0c45ca4968221ec7a', connectivity type 'Public', and state 'Available'.

- Now click Create NAT gateway.

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

The screenshot shows the 'Create NAT gateway' form. The 'NAT gateway settings' section is expanded. The 'Name - optional' field is highlighted with a red box and contains the text 'NAT'. The 'Availability mode' section has two options: 'Regional - new' and 'Zonal'. The 'Zonal' option is selected and highlighted with a blue box. The 'Subnet' dropdown menu is also highlighted with a red box and shows 'subnet-0e7823bb0276394a2 (Public_Subnet_1A)' as the selected option.

- Now Enter Gateway name.
- Availability mode as Zonal.
- Select Public_Subnet_1A
- Now scroll down.

Connectivity type

Select a connectivity type for the NAT gateway.

- ☒ Public
☐ Private

Elastic IP allocation ID [Info](#)

Assign an Elastic IP address to the NAT gateway.

eipalloc-08b23e8abf29d11fb

[Allocate Elastic IP](#)

► **Additional settings** [Info](#)

- Now click Allocate Elastic IP.
- Now scroll down.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

X

Value - optional

Q NAT

X

[Remove](#)

[Add new tag](#)

You can add 49 more tags.

[Cancel](#)

[Create NAT gateway](#)

- Now click Create NAT gateway.
- Now we have to setup this 0.0 towards the net gateway.

▼ Virtual private cloud

Your VPCs

Subnets

[Route tables](#)

Internet gateways

Egress-only Internet gateways

DHCP option sets

NAT gateways (2) [Info](#)

Q Find NAT gateways by attribute or tag



[Actions](#)

[Create NAT gateway](#)

< 1 >



	Name	NAT gateway ID	Connectivity...	State	State message
☐	vpc1-nat	nat-0c45ca4968221ec7a	Public	Available	–
☐	NAT	nat-0b0315b1af281d5f4	Public	Available	–

- Our NAT gateway is created.
- Now we have need to connect the NAT Gateway to our Main Route table.
- For we go inside route table for that click Route tables.

Route tables (1/6) [Info](#)

Last updated
34 minutes ago



[Actions](#)

[Create route table](#)

Q Find route tables by attribute or tag

< 1 >



	Name	Route table ID	Explicit subnet associ...	Edge associations
☒	Main	rtb-0457fe28e5ad5cea7	–	–
☐	RT For Public	rtb-0b72240aa9b00dfea	2 subnets	–
☐	–	rtb-017299725a38d5666	2 subnets	–
☐	–	rtb-0aee4d9128c0e3ce0	–	–

rtb-0457fe28e5ad5cea7 / Main



Details

[Routes](#)

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Both

[Edit routes](#)

Q Filter routes

< 1 >



Destination	Target	Status	Propagated	Route Origin
-------------	--------	--------	------------	--------------

- Now select Main.
- And click Routes and then click Edit routes.

Route 2
Destination

Target

Status
-

Propagated
No

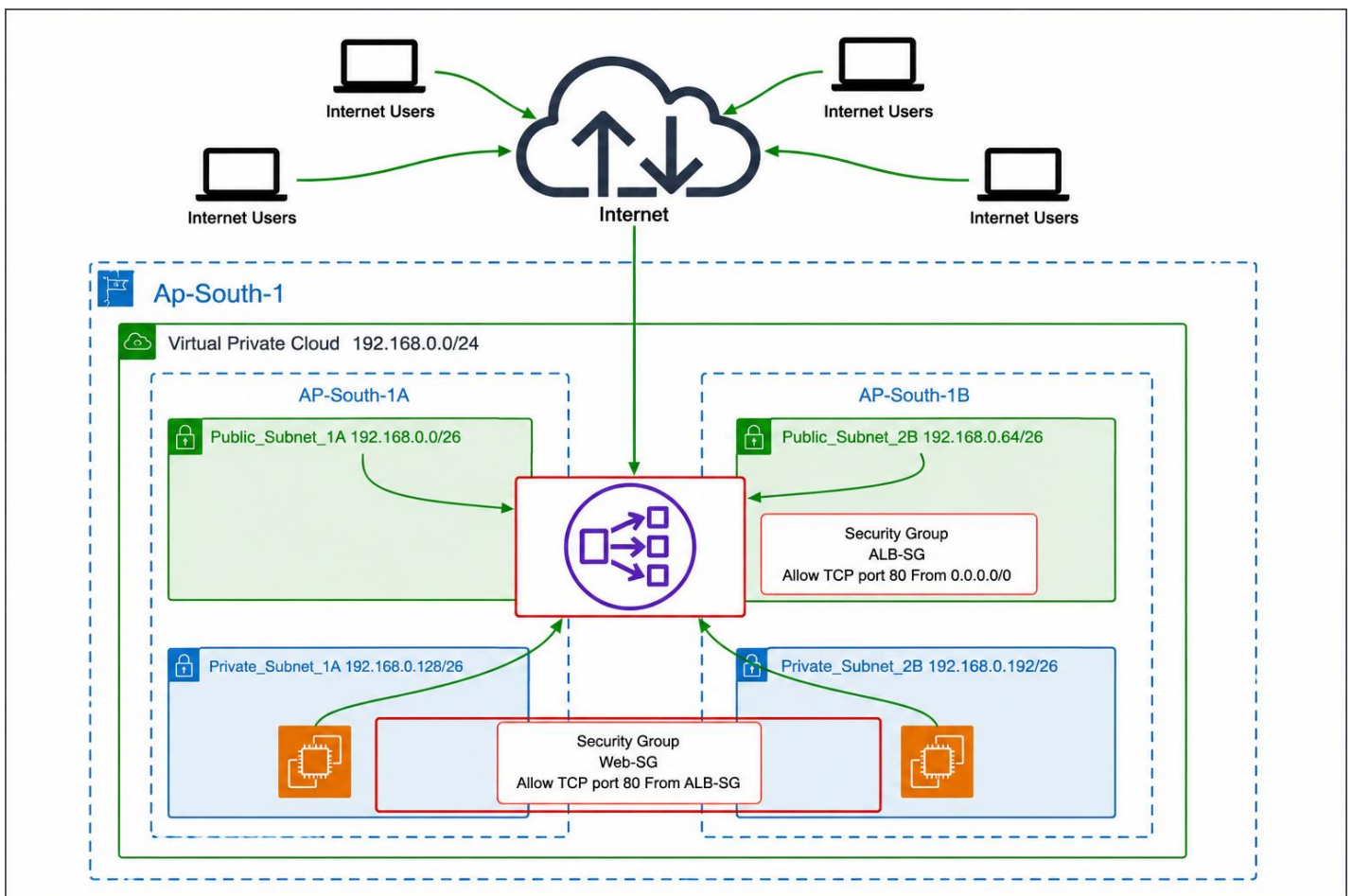
Route Origin
CreateRoute

Remove

Add route

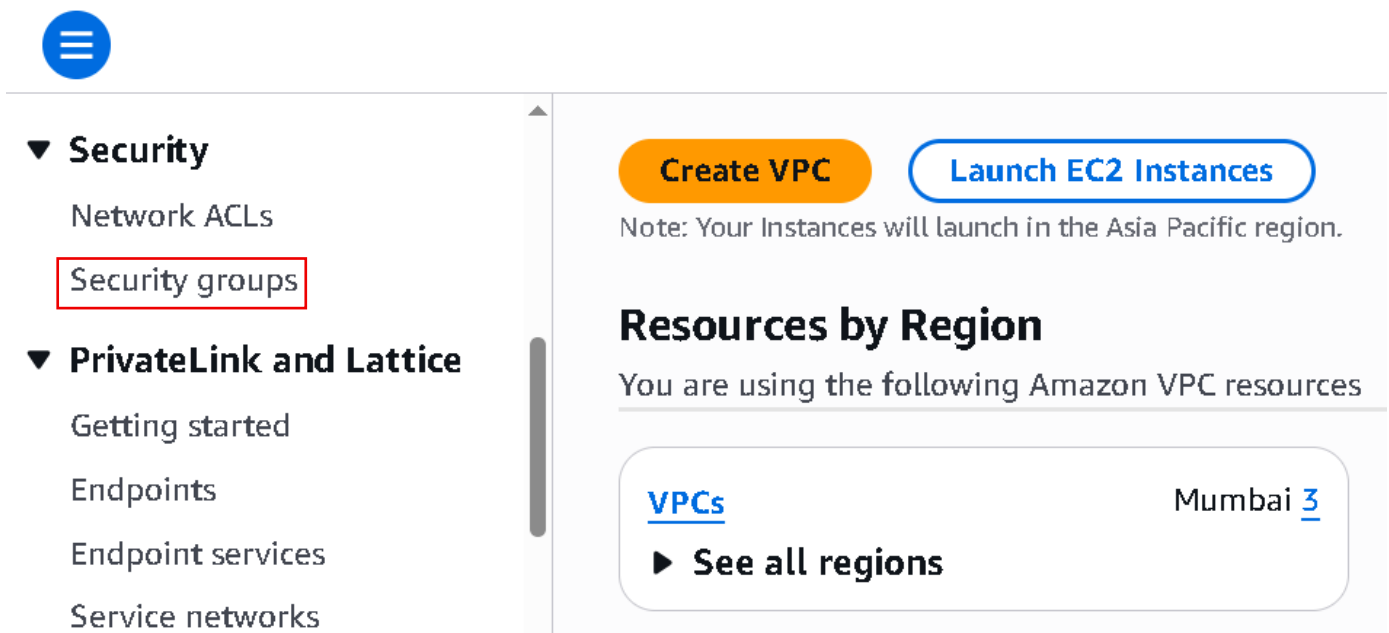
Cancel Preview Save changes

- Now Firstly, click Add route and then Enter 2nd Route details.
- Select Destination 0.0.0.0/0
- Select Target NAT Gateway
- Then select Created NAT Gateway
- And simply click Save Changes.
- And now we are ready to implement load balancer.
- As per rule when you are going to setup your load balancer you have to follow VPC best practices.
- And whatever we are doing right now we are setting up this according to the AWS best practices.



- So, in this image you see users from all over the internet will send requests to my load balancer URL.
- And then my load balancer will send this request to the EC2 instance.

- We have to setup security group at two level.
- The first level we have is to set up a security group for load balancers(Alb-SG).
- And as we know that users over the internet will send requests to your load balancers on a port number 80.
- So, I just created other security group web-SG allows TCP port no 80 from(Alb-SG) anywhere because anybody over the internet can open website.
- And Second level we set up security group for the source(web-SG).
- Because accept the request if the request is coming from only load balancer.
- First we create our Security group then two EC2 instance.
- Then both thing done I create our Network Load balancer.



Security

- Network ACLs
- Security groups**

PrivateLink and Lattice

- Getting started
- Endpoints
- Endpoint services
- Service networks

Create VPC **Launch EC2 Instances**

Note: Your Instances will launch in the Asia Pacific region.

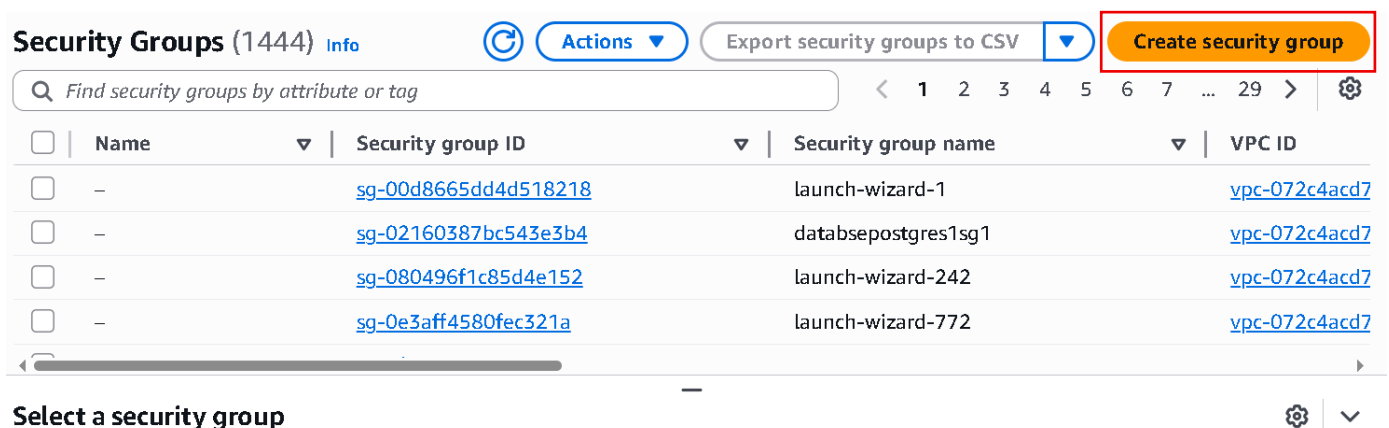
Resources by Region

You are using the following Amazon VPC resources

VPCs Mumbai **3**

See all regions

- Now opens VPC and in Nav bar scroll down.
- And click Security groups in Security.



Security Groups (1444) [Info](#) [Actions](#) [Export security groups to CSV](#) **Create security group**

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID
-	sg-00d8665dd4d518218	launch-wizard-1	vpc-072c4acd7
-	sg-02160387bc543e3b4	databasepostgres1sg1	vpc-072c4acd7
-	sg-080496f1c85d4e152	launch-wizard-242	vpc-072c4acd7
-	sg-0e3aff4580fec321a	launch-wizard-772	vpc-072c4acd7

Select a security group

- Click security groups.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

- Enter security group name (Alb-SG).
- Then Enter description (Securing My ALB From Internet).
- And then select Create VPC.

Inbound rules [Info](#)

This security group has no inbound rules.

[Add rule](#)

- Now click Add rule.

Inbound rules [Info](#)

Type [Info](#)

Protocol

Port range [Info](#)

Source [Info](#)

Description - optional [Info](#)

[Info](#)

[Delete](#)[Add rule](#)

- Now select type as HTTP.
- And Select source From Anywhere.
- Why anywhere because we are going to get request from internet so that anybody can open my website.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags

[Cancel](#)[Create security group](#)

- Now simply scroll down and click Create Security group.

Security Groups (1) [Info](#) [Actions](#) [Export security groups to CSV](#) [Create security group](#)

Find security groups by attribute or tag

Alb-SG [Clear filters](#)

Name	Security group ID	Security group name	VPC ID
-	sg-0b3ee61463ef94613	Alb-SG	vpc-03279cdb55

Select a security group

- First, Security group is created.
- Now we create 2nd Security group.
- So, click create security group.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

web-SG

Name cannot be edited after creation.

Description [Info](#)

Securing my web servers and allowing traffic from ALB only

VPC [Info](#)

vpc-03279cdb550dee663 (MyVPC)

- Enter Security group name (web-SG)
- Enter Description(Securing my web servers and allowing traffic from ALB only).
- Select your Created VPC.

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sg-088aa13bef1d01cc0	HTTP	TCP	80	Cu...		Delete
-	SSH	TCP	22	An...		Delete

[Add rule](#)

- Now add Inbound rules.
- Select type HTTP.
- Then Select Source Alb-SG security group that you created.
- And then add one more inbound rules.
- Select type SSH
- And then source Anywhere.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags

[Cancel](#)

[Create security group](#)

- Now simply scroll down and click Create Security group.
- Now our two Security group is ready,
- Now I am going to create two EC2 instance.

The screenshot shows the AWS Management Console for the EC2 service. The left sidebar contains navigation links for EC2, Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, and Savings Plans. The main content area displays the 'Instances (2)' page. At the top, there are buttons for 'Connect', 'Instance state', 'Actions', and 'Launch instances' (highlighted with a red box). Below these buttons is a search bar and a table of instances. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, and Alarm status. Two instances are listed: 'Private-Instance' and 'Public-Instance1', both in a 'Running' state. Below the table, there is a 'Select an instance' dropdown menu (highlighted with a red box) and a 'Launch instances' button.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
Private-Instance	i-002f9967d6bbb860e	Running	t2.medium	2/2 checks passed	View alarm
Public-Instance1	i-0c0c972f67af41ed2	Running	t2.medium	2/2 checks passed	View alarm

- Now opens EC2 in new tab.
- And click Launch instance.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

web-server-1

[Add additional tags](#)

- Enter instance name (web-server-1).
- Scroll down.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

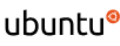
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

Quick Start


Amazon Linux


macOS


Ubuntu


Windows


Red Hat


SUSE


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

- Choose OS in my case I choose Amazon Linux.
- Scroll down

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Linux base pricing: 0.0104 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

- Choose t3.micro this is Free tier eligible.
- Scroll down.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select

 [Create new key pair](#)

- Click Create new key pair.

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

demo

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Cancel

Create key pair

- Enter key name.
- Key pair type as RSA
- And .pem file
- And last click Create key pair
- Now your keypair is downloaded and save it.
- Now scroll down.

▼ Network settings

Info

Edit

Network

Info

vpc-072c4acd761c3b942

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

- Now simple Edit.

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-03279cdb550dee663 (MyVPC)
192.168.0.0/24



Subnet [Info](#)

subnet-0f82985187f3f7b70 Private_Subnet_1A
VPC: vpc-03279cdb550dee663 Owner: 463646775279
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 59 CIDR: 192.168.0.128/26

[Create new subnet](#)

Auto-assign public IP [Info](#)

Disable

- Now select created VPC.
- And then Select subnet private_Subnet_1A.
- And Auto-assign public IP Disable.
- Scroll down.

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups [Info](#)

Select security groups



web-SG sg-0f4faf3ca638a151e ✕
VPC: vpc-03279cdb550dee663

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

- Select Firewall (security group) existing security group.
- And select your created web-SG security group.
- Now we have our EC2 instance in a private subnet okay.
- Now how we will place our website over there because if you want to access this we have to create a EC2 instance connect End point in order to access this.
- Because our EC2 instance is in a private subnet and we don't have public IP.

▼ Advanced details [Info](#)

Domain join directory [Info](#)

Select

 [Create new directory](#) 

IAM instance profile [Info](#)

Select

 [Create new IAM profile](#)

Hostname type [Info](#)

IP name

DNS Hostname [Info](#)

- So, for that scroll down and open Advanced details.
- And scroll down.

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

 [Choose file](#)

```
#!/bin/bash

yum update -y

yum install httpd -y

systemctl start httpd
systemctl enable httpd

echo "Welcome To Web-Server-1" > /var/www/html/index.html
```

- Now here paste the script that provided in your resource.
- Now simply scroll down.

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11.....[read more](#)

ami-0cc96c4cd98401dae

[Cancel](#)

[Launch instance](#)

 [Preview code](#)

- Now leave remaining setting as it is.

- And click Launch instance.
- Now we need to create 2nd EC2 name web-server-2 and same network setting only one change subnet is Private_subnet_2B
- And User data for 2nd EC2 instance is provided in Resource.

EC2 > Instances

Instances (4) Info Last updated 3 minutes ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets Choose filter set

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm state
☐	web-server-2	i-070ab8b80742a8e62	Running	t3.micro	3/3 checks passed	View alarm
☐	ok	i-060297b75b2e06c09	Terminated	t2.medium	-	View alarm
☐	web-server-1	i-0c4152ebcd51c637b	Running	t3.micro	3/3 checks passed	View alarm
☐	linuxServer1	i-04a46245bb334c2c5	Terminated	t3.micro	-	View alarm

- Our both EC2 instance is created.
- Now click Load balancer for creating Network Load Balancer.

Load balancers What's new? [Actions](#) [Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

☐	Name	State	Type	Scheme	IP address type	V

- Now simply click Create load balancer.

EC2 > Load balancers > Compare and select load balancer type

Compare and select load balancer type

A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

Load balancer types

Application Load Balancer Info

Network Load Balancer Info

Gateway Load Balancer Info

- Now we are going to create Network Load Balancer.

- Scroll down and click create.

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.

- Now Enter load balancer name (NLB).
- Scheme Internet-facing.
- Now scroll down.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to on-premises targets or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

 192.168.0.0/24


[Create VPC](#)

- And in Network mapping select your create VPC.
- Now scroll down.

Availability Zones and subnets

Select one or more Availability Zones and corresponding subnets. Enabling multiple Availability Zones increases the fault tolerance of your applications. The load balancer routes traffic to targets in the selected Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

192.168.0.0/26

Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

☒ Assigned by AWS

☐ Use an Elastic IP address

☒ ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

192.168.0.64/26

Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

- Now select Availability zone and subnets.
- Select both Public_Subnet_1A and Public_Subnet_2B.
- IPv4 address Assigned by AWS (means ip address is dynamic).

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups - *recommended*

Security groups support on Network Load Balancers can only be enabled at creation by including at least one security group. You can change security groups after creation. The security groups for your load balancer must allow it to communicate with registered targets on both the listener port and the health check port. For PrivateLink Network Load Balancers, security group rules are enforced on PrivateLink traffic; however, you can turn off inbound rule evaluation after creation within the load balancer's Security tab or using the API.

Select up to 5 security groups



[Create security group](#)

Alb-SG (sg-0b3ee61463ef94613) ✕
Inbound rule (1) Outbound rule (1)

- Now select Created Security group (Alb-SG).

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener TCP:80

Remove

Protocol

TCP

Port

80

1-65535

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#)

Target group

Select a target group



Weight

1

0-999

Percent

100%

- Now protocol TCP and port 80.
- Now we need a target group.
- So, click Create target group.

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - *immutable*

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

☐ IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

☐ Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

☐ Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

- Now select setting Instances.

Target group name

Name must be unique per Region per AWS account.

Target-1

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 8/32

Protocol

Protocol for communication between the load balancer and targets.

TCP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

- Now Enter Name of target group.
- Scroll down.

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#) 

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-03279cdb550dee663 (MyVPC)

192.168.0.0/24



- Now Select the create VPC.
- And simply scroll down.
- And click Next.

Available instances (2/2)

< 1 >

⚙️

☒	Instance ID	Name	State	Security group
☒	i-0a75d28834354cf98	web-server-2	✓ Running	web-SG
☒	i-04f54821d318f2a1b	web-server-1	✓ Running	web-SG

2 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

1-65535 (separate multiple ports with commas)

Include as pending below

- Now select the Available instances that you created.
- And then click include as pending below.

Review targets

Targets (2)

Show only pending

< 1 >

⚙️

Instance ID	Name	Port	State	Security groups	Zone
i-0a75d28834354cf98	web-server-2	80	✓ Running	web-SG	ap-south-1b
i-04f54821d318f2a1b	web-server-1	80	✓ Running	web-SG	ap-south-1a

2 pending

Cancel

Previous

Next

- Now simply click Next.

Step 2: Register targets

[Edit](#)

Targets (2)

Instance ID	Name	Port	Zone
i-0a75d28834354cf98	web-server-2	80	ap-south-1b
i-04f54821d318f2a1b	web-server-1	80	ap-south-1a

[Cancel](#)[Previous](#)[Create target group](#)

- Now simply click Create target group.
- Now our load balancer is created.
- And go back to Load balancer creation tab.

▼ Listener TCP:80

Protocol

TCP

Port

80

1-65535

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group

Target-1

Target type: Instance, IPv4 | Target stickiness: Off

TCP



Weight

1

0-999

Percent

100%

- Now click Refresh and select Target-1.
- Now simply scroll down.

Creation workflow and status

► Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

[Cancel](#)[Create load balancer](#)

- And click Create load balancer.

NLB



Actions ▾

▼ Details

Load balancer type
Network

Status
⋮ Provisioning

VPC
[vpc-03279cdb550dee663](#) ↗

Load balancer IP address type
IPv4

Scheme
Internet-facing

Hosted zone
ZVDDRQ08TROA

Availability Zones
[subnet-0e7823bb0276394a2](#) ↗
ap-south-1a (aps1-az1)
[subnet-0eeb1e9dd70507728](#) ↗
ap-south-1b (aps1-az3)

Date created
May 14, 2026, 13:43
(UTC+05:30)

Load balancer ARN
 `arn:aws:elasticloadbalancing:ap-south-1:463646775279:loadbalancer/net/NLB/a9611435c113c56a`

DNS name info
 `NLB-a9611435c113c56a.elb.ap-south-1.amazonaws.com` (A Record)

- Now NLB is Created you need to wait for its status become active.
- Then go to your target group and open created Target-1.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)



Deregister

Register targets

🔍 Filter targets

< 1 > ⚙️

☐	Instanc...	Name	Port	Zone	Health ...	Health st...	Admini...
☐	i-070ab8...	web-serv...	80	ap-south-...	🟢 Healthy		⊖ No override.
☐	i-0c4152e...	web-serv...	80	ap-south-...	🟢 Healthy		⊖ No override.

- Now simply scroll down and check health status.

NLB



Actions ▾

▼ Details

Load balancer type
Network

Status
🟢 Active

VPC
[vpc-03279cdb550dee663](#) ↗

Load balancer IP address type
IPv4

Scheme
Internet-facing

Hosted zone
ZVDDRQ08TROA

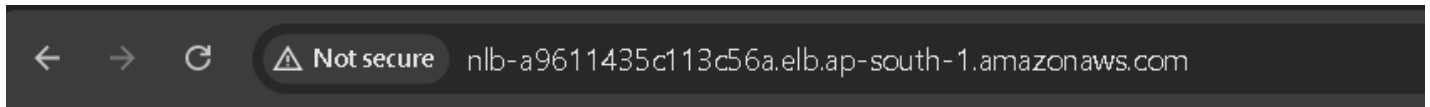
Availability Zones
[subnet-0e7823bb0276394a2](#) ↗
ap-south-1a (aps1-az1)
[subnet-0eeb1e9dd70507728](#) ↗
ap-south-1b (aps1-az3)

Date created
May 14, 2026, 13:43
(UTC+05:30)

Load balancer ARN
 `arn:aws:elasticloadbalancing:ap-south-1:463646775279:loadbalancer/net/NLB/a9611435c113c56a`

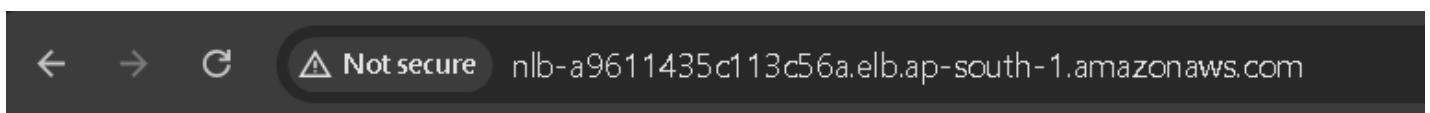
DNS name info
 `NLB-a9611435c113c56a.elb.ap-south-1.amazonaws.com` (A Record)

- Now our Load Balancer status is active now copy the DNS name.
- And open it in new tab.



Welcome To Web-Server-1

- Now you see its working.
- And when you refresh it shift dynamically load on both instance.
- Now refresh this tab.



Welcome To Web-Server-2

- Now you see it's balancing load on both instances.

NOTE: - In Network load Balancer is working with layer four. Because it's working with layer 4 it is having ultra-low latency and it can handle traffic from millions of users, it prefers for gaming servers etc.

Now let's implement cross Zone Load Balancing: -

- It is the most important attribute of load balancer.
- Now open load balancer.

Load balancers (1/1) [What's new?](#)



Actions

Create load balancer



Elastic Load Balancing scales your load balancer capacity automatically in response to changes in demand.

Filter load balancers

☒	Name	State	Type	Scheme
☒	NLB	Active	network	Internet-facing

- Edit IP address type
- Edit subnets
- Manage instances
- Edit health check settings
- Manage listeners
- Edit security groups
- Edit load balancer attributes
- Manage tags
- Delete load balancer

Load balancer: NLB

- Now select Created Load Balancer.
- And click on Action and select Edit load balancer attributes.

Client routing policy (DNS Record)

Applies only to internal requests for clients resolving the load balancer DNS name using [Route 53 Resolver](#).

- ☐ Availability Zone affinity
Client DNS queries will favor load balancer IP addresses in their own Availability Zone. Queries may resolve to other zones if there are no healthy load balancer IP addresses in their own zone.
- ☒ Partial Availability Zone affinity
85% of client DNS queries will favor load balancer IP addresses in their own Availability Zone. The remaining queries will resolve to any zone. Resolving to any zone may also occur if there are no healthy load balancer IP addresses in the client's zone.
- ☐ Any Availability Zone - Default
Client DNS queries will resolve to healthy load balancer IP addresses across all load balancer Availability Zones.

Load balancer targets selection policy

- ☒ Disable cross-zone load balancing - Default
Each load balancer node load balances traffic among healthy targets in its own Availability Zone.
- ☐ Enable cross-zone load balancing
Each load balancer node load balances traffic among healthy targets in all its enabled Availability Zones. [Data transfer charges apply](#)

ARC zonal shift integration

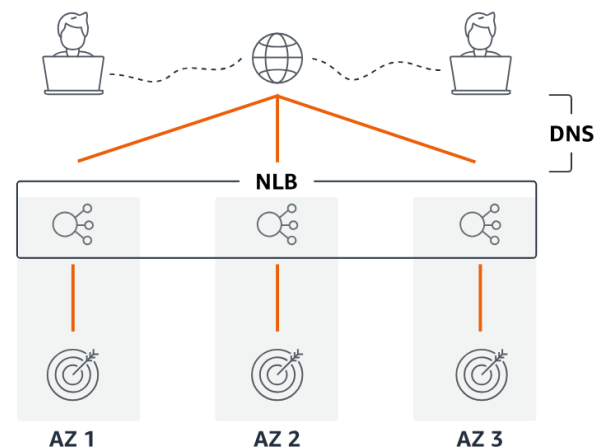
Controls whether Amazon Application Recovery Controller (ARC) zonal shift is available to the load balancer.

- ☒ Disable - Default
Zonal shift will not be available to the load balancer.
- ☐ Enable
Zonal shift will be available to the load balancer.

DNS resolution

☒ Public DNS ☐ Route 53 Resolver

Client routing policy (DNS record) has no effect on public DNS.



- Now, understand when you create a load balancer and select availability Zone in our case we select 2 availability zone south-1a and south-2b in creating load balancer.
- Means in each availability zone one load balancer instance is created.
- Now when you send request to your load balancer this request will be divided into two different node into the two different availability zone when cross -zone is disabled.

Client routing policy (DNS Record)

Applies only to internal requests for clients resolving the load balancer DNS name using [Route 53 Resolver](#).

- ☐ **Availability Zone affinity**
Client DNS queries will favor load balancer IP addresses in their own Availability Zone. Queries may resolve to other zones if there are no healthy load balancer IP addresses in their own zone.
- ☒ **Partial Availability Zone affinity**
85% of client DNS queries will favor load balancer IP addresses in their own Availability Zone. The remaining queries will resolve to any zone. Resolving to any zone may also occur if there are no healthy load balancer IP addresses in the client's zone.
- ☐ **Any Availability Zone - Default**
Client DNS queries will resolve to healthy load balancer IP addresses across all load balancer Availability Zones.

Load balancer targets selection policy

- ☐ **Disable cross-zone load balancing - Default**
Each load balancer node load balances traffic among healthy targets in its own Availability Zone.
- ☒ **Enable cross-zone load balancing**
Each load balancer node load balances traffic among healthy targets in all its enabled Availability Zones. [Data transfer charges apply](#).

ARC zonal shift integration | [Info](#)

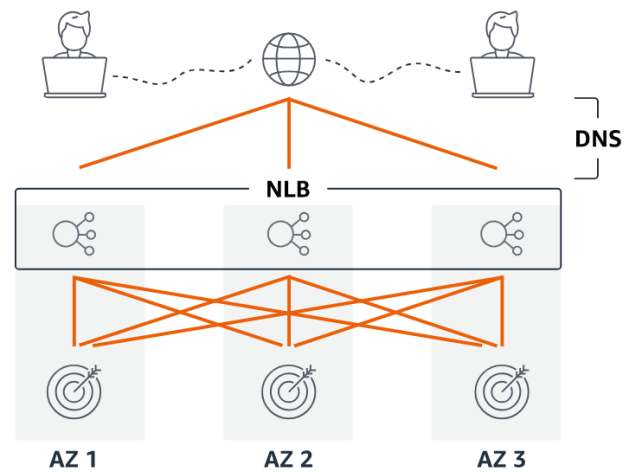
Controls whether Amazon Application Recovery Controller (ARC) zonal shift is available to the load balancer.

- ☒ **Disable - Default**
Zonal shift will not be available to the load balancer.
- ☐ **Enable**
Zonal shift will be available to the load balancer.

DNS resolution

☒ **Public DNS** ☐ **Route 53 Resolver**

Client routing policy (DNS record) has no effect on public DNS.



- When we enable cross-zone load balancing.
- It means any load balancer in node in any availability zone will forward this request to any of the targets.
- It totally depends on you want to enable or disable.